

F O R T I M A N A G E R N S E 6

Lab 10 — High Availability and Device Failure Behaviour

01. FortiManager High Availability

Every lab in this series has run on a single FortiManager. That single appliance now holds the device database, every policy package, the ADOM revisions and the FGFM tunnels to the managed estate — so losing it loses central management of every firewall at once.

FortiManager HA is the answer, and it is a narrower feature than the name suggests. It is redundancy, not load sharing, and the difference matters when sizing an appliance.

► 1.1 The model



Figure 1 — FortiManager HA — one active primary, up to four synchronised secondaries, and the two failover modes.

A secondary does nothing until it is promoted

Only the primary holds FGFM tunnels, accepts configuration changes and runs tasks. A secondary keeps a synchronised copy of the databases and otherwise sits idle.

Adding a secondary therefore buys recovery time, not capacity. An appliance sized at its device limit does not gain headroom by joining a cluster.

► 1.2 The cluster settings page

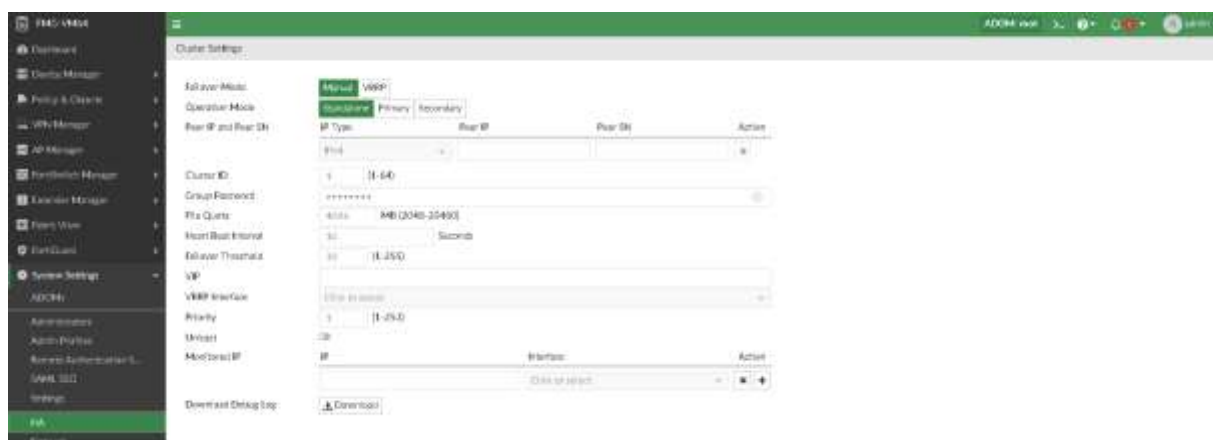


Figure 2 — System Settings > HA — Cluster Settings with the unit in Standalone mode.

Setting	What it controls
Failover Mode	Manual or VRRP. Decides whether a failover requires an administrator and whether devices have to be reointed.
Operation Mode	Standalone, Primary or Secondary. The unit's role in the cluster.
Peer IP and Peer SN	The other cluster members, identified by address and serial number.
Cluster ID and Group Password	Shared membership credentials. They must match on every unit or the cluster will not form.
File Quota	Disk reserved for the files synchronised between members, 2048 to 20480 MB.
Heart Beat Interval	Seconds between heartbeats, 10 by default.
Failover Threshold	Missed heartbeats tolerated before failover, 30 by default — roughly five minutes at the default interval.
VIP, VRRP Interface, Priority	The virtual address, the interface that carries it, and which unit wins when more than one is available.
Unicast	Sends VRRP by unicast instead of multicast, for networks and cloud platforms where multicast is unavailable.
Monitored IP	Addresses the unit pings to decide whether its own path is healthy. Losing them can trigger a failover even when the unit itself is up.

▸ 1.3 Choosing a failover mode

The default heartbeat settings are worth reading together. At a 10-second interval with a threshold of 30, a failed primary is not declared dead for roughly five minutes. That is deliberate — a management plane failing over on a brief network blip is worse than one that waits — but it means FortiManager HA is not a sub-second mechanism and should not be described as one.

Manual failover suits an estate where the devices are few enough to repoint, or where a human decision before promotion is wanted. VRRP suits anything larger, because the alternative is editing central-management settings on every managed firewall while the management plane is down.

02. Simulating a Device Failure

The second half of the lab tests the other direction: not what happens when FortiManager fails, but what FortiManager does when a managed device disappears. The device was taken off the network by shutting the interface FGFM runs over.

```
config system interface
  edit port1
    set status down
  next
end
```

```
fortigate7-2- # config system interface
fortigate7-2- (interface) # edit port1
fortigate7-2- (port1) # set status down
fortigate7-2- (port1) # next
fortigate7-2- (interface) # end
```

Figure 3 — port1 administratively shut on fortigate7-2.

► 2.1 What FortiManager still believes

The device list is checked immediately afterwards.

```
diagnose dvm device list
```

```
diagnose dvm device ...
Connected
FMS-VM64 # diagnose dvm device list
--- There are currently 1 devices/vdoms managed ---
--- There are currently 1 devices/vdoms count for license ---

TYPE      OID      SN      NA      IP      NAME      ADOM      IPS      FIRMWARE      HW GenX
fastigate7-2 165      FGVM64EP-203VB4 - 10.58.228.177 fastigate7-2 root 6.00741 (extended) 7.0 MP2 (1254) N/A
  |~ STATUS: dev=db: not modified; conf: in sync; cond: pending; dm: retrieved; conn: up; template:[unknown]First TEMPLATE
  |~ vdom:[3]root flags:0 adom:root pkg:[modified]fastigate7-2-after-Conflict fmsprof:[modified]template cli:[unknown]try=
Metadata-Object

--- There are currently 0 FortiAP managed ---

--- There are currently 0 FortiSwitch managed ---

--- There are currently 0 FortiExtender managed ---

--- End device list ---
```

Figure 4 — diagnose dvm device list — the device still reported with conn: up.

Observation

The device is unreachable, and the output still reads conn: up with cond: pending.

Why

The connection state is a record of the last known outcome, not a live probe. Nothing has tried to reach the device since the link went down, so nothing has failed yet.

Reading

cond: pending is the only hint that something is unresolved, and it is easy to miss next to a green conn: up.

The FGFM session list agrees, and is equally stale — the tunnel is still listed with an uptime that keeps counting.

```
diagnose fgfm session-list
diagnose debug application fgfm 255
```

```

Connected
FMS-VM64 # diagnose fgfm session-list
fortigate7-2 (165) sn(FGVMEV682-2D3VB4) ip(10.58.228.177)
state(tunnel) tunnel(169.254.0.2) uptime:Thu Oct 8 19:53:30 2026
Session count = 1 (tunnel 1)

FMS-VM64 # diagnose debug application fgfm 255
fgfired debug filter: disable
FMS-VM64 #

```

Figure 5 — diagnose fgfm session-list — the tunnel still present after the device went away.

The same output is also a useful reference for a healthy tunnel: the device name, serial, address, tunnel state, the internal 169.254.0.x tunnel address and the session count.

► 2.2 The operation that exposes it

Nothing reveals the failure until FortiManager is asked to do something with the device. An Update Device was issued, and it sat at 5% with the task pending.

The screenshot shows the 'Update Device' window in FortiManager. At the top, a progress bar is at 5%. Below it, a summary line reads 'Total: 0/1, Pending: 1, In Progress: 0, Completed: 0'. A 'Show Details' toggle is on the right. Below this is a 'View Progress Report' button and a search bar. A table displays the task details:

#	Name	Time Used	Status
1	fortigate7-2	N/A	Pending

Figure 6 — Update Device pending — waiting on a device that is not there.

Two attempts eventually timed out and were recorded as errors.

ID	Source	Description	User	Status	Time Used	ADOM	Start Time	End Time
74	Device Manager	Update Device	admin	Error	1m 0s	root	Thu, Oct 8, 2026 8:48 PM PDT	Thu, Oct 8, 2026 8:49 PM PDT
73	Device Manager	Update Device	admin	Error	4m 13s	root	Thu, Oct 8, 2026 8:45 PM PDT	Thu, Oct 8, 2026 8:49 PM PDT

Figure 7 — Task Monitor — two Update Device tasks ending in Error after 1m 0s and 4m 13s.

2.3 The sequence

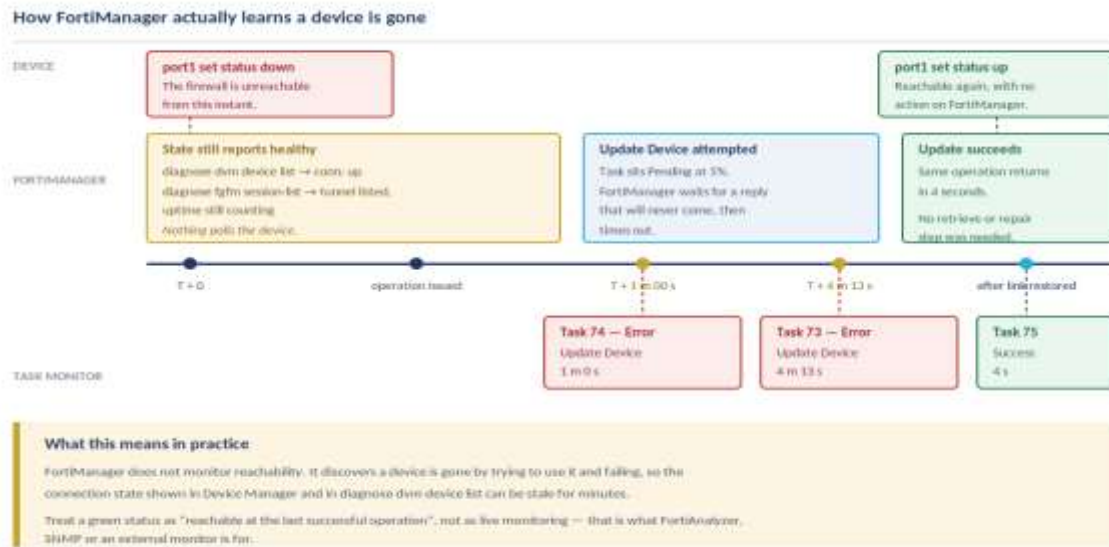


Figure 8 — The full detection sequence, from link down to recovery.

A green status is a memory, not a measurement

FortiManager discovers that a device is gone by trying to use it and failing. Until something is attempted, Device Manager and diagnose dvm device list can report a dead firewall as connected for minutes.

This is the practical limit of FortiManager as a monitoring tool, and the reason availability monitoring belongs to FortiAnalyzer, SNMP or an external monitor rather than to the management plane.

03. Recovery

The interface was brought back up on the device, with nothing done on FortiManager.

```
config system interface
  edit port1
    set status up
  next
end
```

```

fortigate7-2- # config system interface
fortigate7-2- (interface) # edit port1
fortigate7-2- (port1) # set status up
fortigate7-2- (port1) # next
fortigate7-2- (interface) # end

```

Figure 9 — port1 brought back up.

The same operation that had been failing now completes in four seconds.

The screenshot shows the 'Update Device' window with a progress bar at 100%. Below the progress bar, it says 'Total: 1/1, Success: 1, Warning: 0, Error: 0'. There is a 'View Progress Report' button and a search bar. A table lists the update results:

#	Name	Time Used	Status
1	fortigate7-2	4s	Successfully updated device information.

Figure 10 — Update Device — successfully updated device information in 4s.

ID	Source	Description	User	Status	Time Used	ADOM	Start Time	End Time
73	Device Manager	Update Device	admin	Success: 1	4s	root	Thu, Oct 8, 2026 8:50 PM PDT	Thu, Oct 8, 2026 8:50 PM PDT
74	Device Manager	Update Device	admin	Error: 1	1m 0s	root	Thu, Oct 8, 2026 8:48 PM PDT	Thu, Oct 8, 2026 8:49 PM PDT
73	Device Manager	Update Device	admin	Error: 1	4m 53s	root	Thu, Oct 8, 2026 8:45 PM PDT	Thu, Oct 8, 2026 8:49 PM PDT

Figure 11 — Task Monitor — the successful task recorded above the two failures.

Observation	No retrieve, no re-authorization and no reinstall were needed. The tunnel re-established itself and the device returned to normal management.
Why	FGFM is device-initiated here, so the FortiGate reconnects on its own once the path is back. The registration and certificate state from Lab 04 survive a link outage untouched.
Contrast	This is the opposite of the Conflict in Lab 08, where the device and the database genuinely disagreed. An unreachable device is not an out-of-sync device, and the two should not be treated the same way.

▸ 3.1 What the device record carries

The device list output in section 02 also shows how much state has accumulated on this one device across the series. Alongside the serial, address and firmware, the record names the policy package, the provisioning template, the firmware template and the CLI template assigned to it.

The bracketed state against each one — modified or unknown — is the same vocabulary as the configuration status field, applied per assigned object rather than to the device as a whole.